SE-Lab-7

Fuzzing for Input Validation Bugs

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Repo: L6_L7_05_blood_donation_screening

1. Run the buggy application .py in the repo assigned to you (under Lab7):

`Command: python processor.py`

```
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ cp processor.py processor_buggy_backup.py
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): O+
Validated: O Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/80
BP Metrics: Diff: 40, Ratio: 1.5

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D101|65|13.5
Status: Donor D101: Eligible
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): AB+
Validated: AB Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/0
Traceback (most recent call last):
  File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 50, in <module>
    main()
  File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 43, in main
    print("BP Metrics:", calculate_pulse_pressure(bp_input))
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 21, in calculate_pulse_pressure
    ratio = round(systolic / diastolic, 2)
                  ~~~~~~~~~^~~~~~~~~~
ZeroDivisionError: division by zero
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$
```

Valid input works; BP input '120/0' crashes the buggy script with ZeroDivisionError.

2. Run the test_processor.py:

Command: python -m pytest test_processor.py

```
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python -m pytest test_processor.py
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7
plugins: hypothesis-6.170.15
collected 5 items

test_processor.py FFFF.                                                  [100%]

=================================== FAILURES ===================================
_____________________ test_validate_blood_type_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py", line 6, in test_validate_blood_type_no_crash
  |     def test_validate_blood_type_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py", line 8, in test_validate_blood_type_no_crash
    |     result = validate_blood_type(s)
    |              ^^^^^^^^^^^^^^^^^^^^^
    |   File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 10, in validate_blood_type
    |     return f"{antigen} {rh_map[rh_char]}"
    |                        ~~~~~~^^^^^^^^^^
    | KeyError: '0'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py", line 11, in test_validate_blood_type_no_crash
    |     assert False, f"validate_blood_type crashed with: {type(e).__name__}: {e}"
    | AssertionError: validate_blood_type crashed with: KeyError: '0'
    | assert False
    | Failing test case: test_validate_blood_type_no_crash(
    |     s='0',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py:10
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py", line 8, in test_validate_blood_type_no_crash
    |     result = validate_blood_type(s)
    |              ^^^^^^^^^^^^^^^^^^^^^
    |   File "/home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 8, in validate_blood_type
    |     rh_char = cleaned[-1]
    |               ~~~~~~~^^^^
    | IndexError: string index out of range
    |
```

Hypothesis finds failing inputs, e.g. validate_blood_type('0') raises KeyError.

```
        Assumes string format 'systolic/diastolic' (e.g. '120/80').
        """
        sys_str, dia_str = bp_string.split('/')
        systolic = int(sys_str)
        diastolic = int(dia_str)
        pulse_diff = systolic - diastolic
>       ratio = round(systolic / diastolic, 2)
                      ^^^^^^^^^^^^^^^^^^^^
E       ZeroDivisionError: division by zero

processor.py:21: ZeroDivisionError

During handling of the above exception, another exception occurred:

    @given(st.integers(min_value=-1000, max_value=1000),
>          st.integers(min_value=-1000, max_value=1000))
                ^^^

test_processor.py:34:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

sys_val = 0, dia_val = 0

    @given(st.integers(min_value=-1000, max_value=1000),
           st.integers(min_value=-1000, max_value=1000))
    def test_calculate_pulse_pressure_numeric(sys_val, dia_val):
        try:
            result = calculate_pulse_pressure(f"{sys_val}/{dia_val}")
            assert isinstance(result, str)
        except Exception as e:
>           assert False, f"calculate_pulse_pressure crashed on '{sys_val}/{dia_val}': {type(e).__name__}: {e}"
E           AssertionError: calculate_pulse_pressure crashed on '0/0': ZeroDivisionError: division by zero
E           assert False
E           Failing test case: test_calculate_pulse_pressure_numeric(
E               sys_val=0,  # or any other generated value
E               dia_val=0,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   /home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py:39

test_processor.py:40: AssertionError
=========================== short test summary info ============================
FAILED test_processor.py::test_validate_blood_type_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_pulse_pressure_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_parse_donor_registry_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_pulse_pressure_numeric - AssertionError: calculate_pulse_pressure crashed on '0/0': ZeroDivisionError: division by zero
========================= 4 failed, 1 passed in 3.98s ==========================
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$
```

Short test summary: 4 failed, 1 passed (buggy processor.py).

3. Fix the buggy application python script:

Run the fixed processor.py – Command: python processor.py

```
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ gedit processor.py
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): O+
Validated: O Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/80
BP Metrics: Diff: 40, Ratio: 1.5

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D101|65|13.5
Status: Donor D101: Eligible
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): A+
Validated: A Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/0
BP Metrics: Invalid BP Values

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D102|abc
Status: Invalid Record Format
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): X
Validated: Invalid Blood Type

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/0
BP Metrics: Invalid BP Values

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D102|abc
Status: Invalid Record Format
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$
```

Fixed script handles invalid inputs (X, 120/0, D102|abc) with messages instead of crashing.

4. Re-Run the test_processor.py wrt to fixed processor:

Command: python -m pytest test_processor.py -v

```
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$ python -m pytest test_processor.py -v
============================================================================= test session starts ==============================================================================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- /home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7/venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /home/PES1UG24CS554/L6_L7_05_blood_donation_screening/Lab 7
plugins: hypothesis-6.170.15
collected 5 items

test_processor.py::test_validate_blood_type_no_crash PASSED                                                                                                          [ 20%]
test_processor.py::test_calculate_pulse_pressure_safe PASSED                                                                                                         [ 40%]
test_processor.py::test_parse_donor_registry_safe PASSED                                                                                                             [ 60%]
test_processor.py::test_calculate_pulse_pressure_numeric PASSED                                                                                                      [ 80%]
test_processor.py::test_parse_donor_registry_structured PASSED                                                                                                       [100%]

============================================================================== 5 passed in 0.61s ===============================================================================
(venv) PES1UG24CS554@node2:~/L6_L7_05_blood_donation_screening/Lab 7$
```

All 5 Hypothesis fuzz tests pass on the fixed processor.py.